RAILWAY BOOKING SYSTEM

CO2210 Programming Quest - 3

TEAM MEMBERS :

24/ENG/042

24/ENG/043

24/ENG/035

PRACTICAL DATE : 2026/10/30

SUBMISSION DATE : 2026/10/30

1. INTRODUCTION AND OBJECTIVES

The Sri Lanka Railway Booking Management System is a console based Java application, which is designed for railway ticketing officers. The software assists the officer to perform tasks related to booking of train seats through a menu. The program has been designed for CO2210 Programming Quest 03.

This booking management system is specifically designed for express-train on three routes; Colombo-Kankesanthurai, Colombo-Galle, and Colombo-Kandy. An officer can see train timing, available seats, book, find, amend, and delete bookings respectively. Additionally, the program will maintain details about the train and booking in CSV files, which could be used next time when the program is executed. Payments are not part of this system.

Objectives of the project include:

- Practicing Java programming and OOP
- Using file operations to write and read booking details
- Avoiding overbooking and maintaining four-seats limit for each customer
- Developing a user-friendly system for ticketing officers.

2. PROGRAM DESIGN

The program follows an object-oriented design. It is divided into five classes. Each class has a separate task, which makes the code easier to read, understand, and update. The Main class handles the menu and user input. BookingManager carries out the booking rules, while FileManager works with CSV files.

Class	Purpose in the Program
Main	Displays the menu, reads user inputs, shows results and handles invalid inputs.
Train	Stores the train ID, name, route, starting and ending stations, departure time, arrival time, seat capacity and service days.
Booking	Stores one reservation, including the booking ID, train, travel date, customer details, number of seats and booking status.
BookingManager	Checks train details and seat availability. It creates, searches, changes and cancels bookings.
FileManager	Reads timetable and booking records from CSV files and saves updated bookings.

2.1. Interaction of classes

Upon selection of one of the options from the menu, the function chosen is called from within the class Main and it in turn invokes the appropriate function in the BookingManager class. BookingManager employs Train and Booking objects for manipulation of data related to trains and bookings respectively, and also FileManager class for saving the modified data to the files.

The UML diagram of classes and their interactions is shown in the Class_Diagram.pdf

.3. FILE DESIGN

The application uses two Comma Separated Values (CSV) files inside the data folder. CSV files are easy to read and edit, so a database is not needed for this lab

File	Data Stored
data/trains.csv	Train ID, name, route, starting station, destination, departure time, arrival time, seating capacity and service days.
data/bookings.csv	Booking ID, train ID, travel date, customer ID, customer name, contact number, reserved seats and booking status.

The timetable data file holds sample services for the selected three routes. Bookings data makes use of booking ID that includes B0001. Status is set to be active for an ongoing booking while cancelled when it is cancelled.

The existing train and booking data are read at the start of the program. Whenever a new booking is created, modified, or cancelled, the booking data is stored back to ensure no data is lost whenever the program ends.

4. BOOKING LOGIC

4.1. View trains and check available seats

The officer can view the train timetable and select a train by its ID. The officer then enters a travel date to check seat availability. Available seats are calculated by subtracting the seats in active bookings from the total number of seats for that train and date. Cancelled bookings are not counted.

4.2. Create booking

In order to create a booking, the officer needs to select a train, enter the date, and enter the customer ID number, customer name, customer contact number, and the necessary number of seats. Then the system verifies whether the travel date is appropriate, whether the customer does not surpass the four-seats limit, and whether there are enough seats on the train. If all this information is appropriate, a new booking ID is created and stored in the CSV file. In case if there is no place available, the booking will be declined.

4.3. Amend booking

In order to amend a booking, the officer should enter the booking ID and find the booking in the list of bookings. In case if the booking is active, the officer may change the train, the date of the journey, the number of seats or customer's information.

4.4. Cancellation of booking

The officer enters the booking ID and confirms the cancellation. The booking status is changed to CANCELLED. The reserved seats are made available for other customers. The booking that has been canceled is recorded in the CSV file.

4.5. Search for booking information

A search can be conducted to retrieve information about a particular booking using the booking ID. An officer can also choose a route and train number to retrieve all bookings made or those made on a selected date.

5. ASSUMPTIONS AND LIMITATIONS

The assumptions made for simplicity in the design of the lab program are:

1. There is a single computer terminal for use by the railway ticketing officer.

2. Only the three routes referred to in the assignment are taken into account.

3. The sample timetable involves six trains per day where one train runs on each route on a particular direction.

4. The sample timetable given in the design report contains 48 seats in each train. The values are for the lab purpose and can be altered in trains.csv.

5. The customer can book up to 4 seats for the same train and date. The limit holds even if the customer has multiple bookings on his/her account.

6. The customer has to give his/her ID number, name, and contact number.

7. The booking window can be done within a maximum period of 30 days only, where the booking has to be done a minimum of two hours prior to departure as mentioned in the given design report.

8. Only seat counts are stored. No seat numbers are assigned or any classes of tickets are allotted.

9. Sample timings and train numbers are not real-time rail data. It is assumed that the time zone of the computer is in accordance with Sri Lankan time.

10. The software doesn't contain features such as payments, officer login, internet usage, or simultaneous booking by various officers.

11. User text inputs don't contain any commas or double quotes since simple CSV files are considered.

6. FUNCTIONAL TEST SCENARIOS

The following test scenarios are listed in the supplied design report. The reported outcomes are shown below. They should be checked again using the final Java files and CSV data before submission.

Test Scenario	Expected Results	Reported Results
Create a booking for 3 seats	Available seats decrease by 3.	Passed
Try to reserve more than 4 seats for the same customer, train and date	The booking is rejected.	Passed
Change a booking from 3 seats to 4 seats	Seat availability is updated correctly.	Passed
Cancel an active booking	The reserved seats become available and the record is kept.	Passed
Close and restart the program	Previously saved CSV booking data is still available.	Passed
Fill all seats and request one more seat	The extra booking is rejected.	Passed
Use the menu to create, find, modify, view and cancel a booking	Each menu option performs the expected task.	Passed

7. POSSIBLE FUTURE IMPROVEMENTS

The current system can be considered to be an appropriate one for lab demostration purpose. The system may be improved as follows:

- Live timetable details: Integrate the system with live train timetables from the website of Sri Lanka Railways instead of using the sample data.
- Database: Implement MySQL or PostgreSQL databases to maintain safety and to accommodate numerous ticket officers.
- Login of officer: Introduce username-password and access control for officers.
- Seats and tickets: Allow the officers to reserve seats and choose their class.
- Multiple route selection: Implement multiple route selection, including stations, train and non-train operating days.
- Ticket and notifications: Develop printing ticket facility and allow sending booking details via SMS and/or email.
- Backup and History: Maintain history log and backup/restore facilities.
- Online Booking and Payments: This can be included in a future release, but payment processing is out of scope for this assignment.

In summary, this Java application fulfills all the requirements of this lab through object-oriented classes, a console menu and the use of CSV files for handling bookings on railways. The basic booking policies enable officers to make seat bookings without exceeding the capacity.

8. References

1. Department of Computer Engineering, University of Sri Jayewardenepura. CO2210 Programming Quest - Assignment 03 (lab assignment sheet).

2. Sri Lanka Railways. Official website: https://railway.gov.lk/ (website referenced by the assignment).

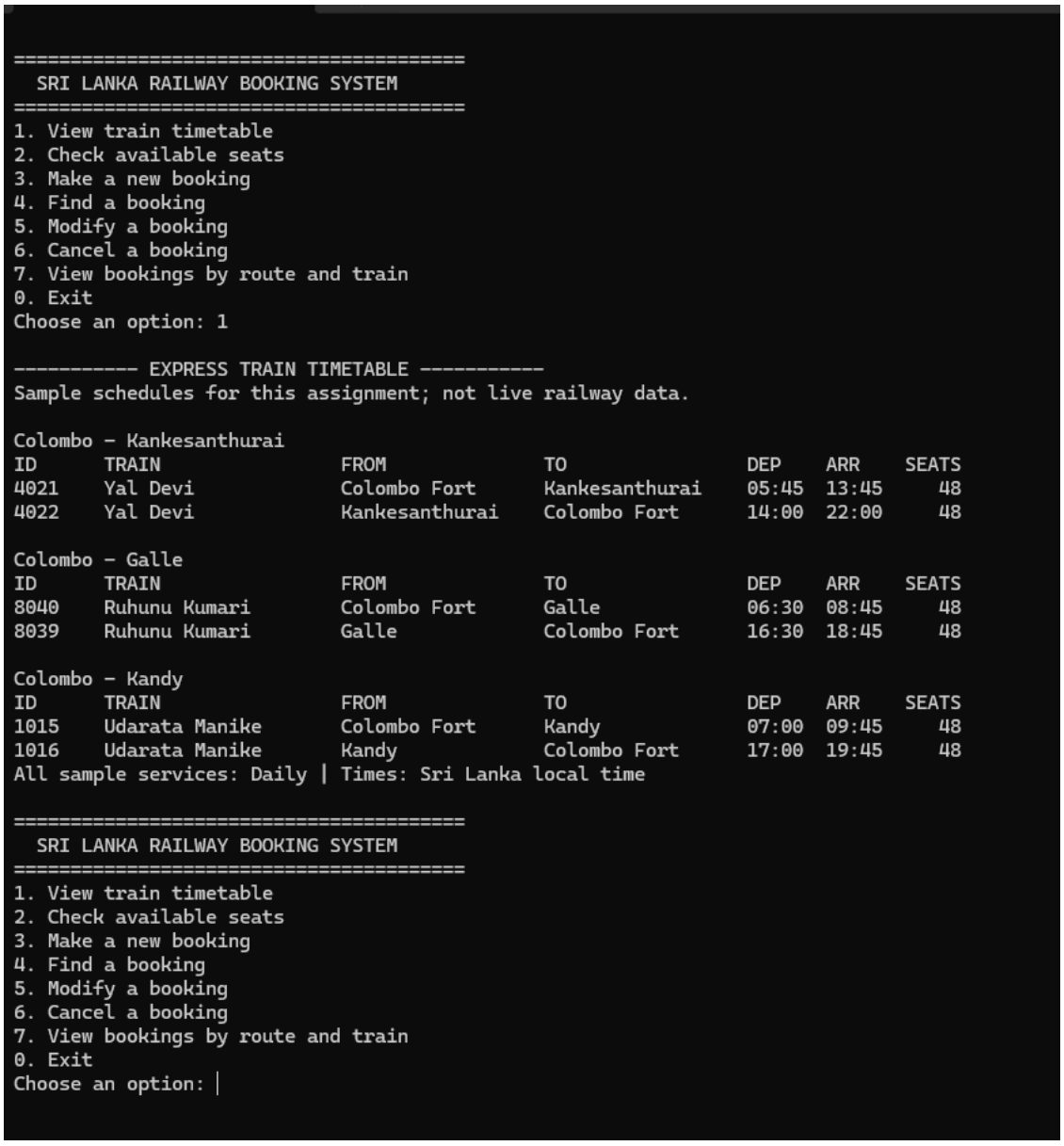


FIGURE 1 : BOOKING SYSTEM UI